This is Chemistry 3A

If you are not enrolled or waitlisted, see your counselor

All others will sign in with Attendance Code during this first day in lecture

- **Seats are available first to enrolled students**
- **Waitlisted students will have to stand or find useful floor space for sitting if no available seating**

Chemistry 3A

Introductory General Chemistry

Course Administration & Policies

S. M. Halloran

Fresno City College

Course Facts & Information

Know your section number

There are 30 enrolled students in each section

Both sections, 60 students total, will be here in room NS-101 (yes, here) from 5:00-6:20 p.m. on Tuesday and Thursday

Right after lecture, one section will have its lab sessin in room NS-339 starting at 6:30 pm

Waitlisted Procedure

- Only students on waitlist have a chance of being added to course
- Enrolled students who are no-shows in 1st lecture get 24 hours to make contact with instructor if they wish to continue enrollment
- Waitlisted students attending 1st lecture & submitting Attendance Code will be given Add Authorization IF slots become available and in order of the queue of intent-to-enroll on waitlist. A message will be sent to add the course

  Students on waitlist given Add Authorization have 24 hours to add the course, or the Add Authorization is removed & given to next student on waitlist
- The last day a waitlisted student is able to add is the 1st day of the 3rd week of semester

Your Instructor

Mitch Halloran

- "Mitch"
- "Mister M", "Mister H", "Mister Mitch"
- "Dr. H", "Dr. Mitch", "Dr. M"

Probably not

- "Professor Halloran" (but thank you)
- "Mitchell" (certainly not! Not even Mom called me that except in a fit)

The Project

- To see you realize the objectives of the Chemistry 3A course in the Department of Chemistry at Fresno City College
- In 18 weeks
- Through implementation of the course design already developed

Communication Plan

- All projects—like your success in this course—write out a communication plan
- This is about what and how to communicate to move the project along
- What to communicate
  - Asking about how to solve a problem, understanding details of an assignment, when is a deadline
  - Helping each other with reminders, setting up an interactive meeting online or in person
- **How to communicate**
  - **Canvas Inbox/Messages**

    Avoid “reply”ing to Canvas Announcements, Assignments, etc. Might not be seen
  - Email: mitch.halloran@fresnocitycollege.edu
  - **Zoom**: “online office hours” (MORE TO FOLLOW)

Working Successfully in Chem 3A

Syllabus

- Answers questions about objectives, content, structure of the course with a broad overview and outline of the course
- Information about required materials for the course
- freely accessible online textbook
- lab gear (coat + goggles)
- scientific-grade calculator for use on examinations
- Grading: components of performance in course
  - Examinations: midterms and final
  - Reports of laboratory experiments submitted
  - Homework: pre-lecture and pre-lab quizzes, graded assignments, ungraded offerings (practice exams)
- Policy on late work submission, missed exams (retakes), etc

Working Successfully in Chem 3A

Online Systems/Tools

- Canvas
  - Modules – each week a module will be published and you should find your links to what you need to do to doing reading, complete quizzes, find homework
  - Announcements – look for this regularly
  - Course Home Page

Interaction between instructor & student(s)

  - Part-time instructors interact through Canvas and email

Email: Mitch.Halloran@fresnocitycollege.edu

CHEM-3A-19320-2026SP

Spring 2026

- Home
- Announcements
- Modules
- Assignments
- Zoom
- Grades
- Quizzes
- Files
- Simple Syllabus
- Pages
- Discussions
- People
- BigBlueButton
- Collaborations
- FCC Library
- Office 365
- FCC Tutorial Services
- Google Drive
- Net-Tutor (24/7) Virtual Tutoring
- Pisces
- Virtual Hub for Students
- Chat
- Lucid (Whiteboard)
- Starfish
- Parchment Badges
- NameCoach
- My Bookstore

Canvas Features

- Modules your main go-to
- Announcements important
- Grades!
- Zoom – your walk-in "office hours"

- Starfish – helps instructor contact with those who can help you deal with college
- Conference : BigBlueButton – works like Zoom?
- NameCoach: help me & others pronounce your name
- Chat – can be a thing
- Lucid – need to work with that

Working Successfully in Chem 3A

Attendance

- Being at lecture & certainly being in the lab is not optional: attendance is recorded
- While attendance is not graded, missing lectures can result in being dropped from course, getting messages from instructor
- Missed laboratory sessions get a ZERO for required report of that experiment

Time Commitment

So for a 4-unit course:

- **Class time (in-person or online):** ~4 hours per week
- **Out-of-class study time:** ~8–12 hours per week
- **Total estimated time commitment: 12–16 hours per week**

ChatGPT estimate

Rubrics

- A **rubric** is a structured scoring guide that defines the criteria, performance levels, and expectations used to evaluate the quality of a student's work
- It clarifies what matters, how it will be judged, and what distinguishes stronger from weaker performance.

- Rubrics in **Canvas** are to be used for
  - laboratory report submissions
  - homework
- Students should look at the rubric so they can see what is expected of them in the submission of work, and BEFORE the work is submitted

Content For This Course

All of the content of this course, including the PowerPoints of these lectures (in PDF form), should be found in some place on the Canvas app

And if you don’t find what you are looking for, send me a message right away

Artificial Intelligence in Learning & Academics

- The elephant in the room? Have you been talking about it in your course of learning?
- Will be transformative, but…
  - useful to you IF you use it properly
  - ChatGPT/Bing CoPilot/Gemini don’t initiate the “what if” and “why” that drives discovery and exploration
  - A.I. does not and will not replace humans in asking questions that push discovery
  - Learning the foundations of a discipline—that is, getting that Associates or Bachelors or Doctoral degree—makes you that better thinker who is asking “what if” and “why”
- Doing that “homework”…
  - Helps you pass the test, yes
  - But also builds you as a discoverer, explorer, researcher, thinker

Artificial Intelligence in Learning & Academics

This is why you take Chem 3A & learn chemistry: to prepare you for doing what A.I. can't [ever] do

Analysis of Patterns & Signals, Information & Knowledge Recall

Discovery, Exploration, Inquisitiveness/ Curiosity, Research

A.I. → **Typically faster**

YOU → **Within human limitations** → **No limitations to the potential of the mind that made A.I.**

Artificial Intelligence in Learning & Academics

🧠 Humans vs 🤖 AI — Capabilities and Limits

Capability	Human Mind	AI
Understanding	Builds deep conceptual frameworks; connects abstract ideas to real-world experiences; can adapt knowledge to new, unfamiliar situations.	Recognizes patterns and applies known methods quickly, but doesn't truly "understand" concepts — it works from correlations in data, not meaning.
Reasoning Across Contexts	Can integrate knowledge from unrelated fields (e.g., chemistry + ethics + economics) to solve novel problems.	Works well when the problem is structured and similar to training data; struggles with integrating concepts from vastly different domains in creative ways.
Error Detection	Can notice when an answer "doesn't make sense" physically, chemically, or ethically, even without knowing the exact answer.	May produce a perfect-looking but fundamentally wrong answer ("hallucination"), without any internal alarm.
Creativity & Insight	Generates original hypotheses, finds elegant shortcuts, invents new problem-solving methods.	Can recombine known ideas in novel ways, but doesn't invent *truly new* scientific concepts without human framing.
Learning from Experience	Learns continuously from mistakes, success, and direct sensory experience.	Cannot self-learn from direct real-world experiences — only retrains when humans feed it new data.
Ethics & Judgment	Weighs social consequences, moral implications, and values in decisions.	Lacks intrinsic moral understanding — applies rules only as programmed.
Motivation	Has curiosity, drive, and personal goals — can choose to explore beyond requirements.	Has no self-motivation; runs tasks only when prompted.